

KARIM SOUISSI

Graduate AI Engineer — Data Science & Decision Systems

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PROFILE

Fresh graduate AI Engineer with hands-on experience in LLMs, NLP, multi-agent systems, and machine learning, gained through industry internships and applied projects. Passionate about building intelligent, production-ready AI systems that deliver real-world impact.

EDUCATION

Engineering Degree, Data & Decision Systems — ENET'Com, Sfax	2023 – 2026
National School of Electronics and Telecommunications of Sfax — Option: Data Science (AI Engineering)	
Physics–Technology Preparatory Cycle — IPEIS, Sfax	2021 – 2023
Preparatory Institute for Engineering Studies — national exam rank 128/800	
Scientific Baccalaureate, Technical Sciences — High School 9 Avril, Sfax	2017 – 2021
Grade: 17.02/20	

PROFESSIONAL EXPERIENCE

AI Engineer Intern — DataOps Copilot (End-of-Studies Project) — KPIT Engineering	Feb 2026 – Present
- Built an autonomous multi-agent system (LangGraph) performing end-to-end root cause analysis of Jenkins data-pipeline failures, coordinating RCA, Investigator, and Remediation agents through a stateful, conditionally-routed workflow with self-correction. - Implemented MCP (Model Context Protocol) integration to enable controlled access to internal repositories, configuration files, and SQL Server data sources in an enterprise environment. - Engineered a two-layer log-parsing mechanism compressing raw Jenkins logs by up to 83% (token count), plus a retrieval-augmented memory layer (FAISS golden-case store) enabling the system to learn from past incidents. - Added a human-in-the-loop feedback loop and an evaluation harness (LLM-as-a-judge, latency and cost metrics). - Shipped Terminal and React UIs with live agent-reasoning streaming, backed by a Flask API.	
AI Engineer Intern — Agentic Code Intelligence System — KPIT Engineering	Jun – Aug 2025
- Built an agentic AI system for enterprise-scale code understanding and pull request analysis, grounded in Jira ticket context, using the ReAct reasoning framework. - Integrated AST-based code parsing, RAG pipelines, Neo4j knowledge graphs, Chroma DB vector database, and LangGraph to enable autonomous code analysis and context-aware reasoning across enterprise repositories.	
Data Engineer Intern — Data Pipeline Monitoring & Alerting System — PRIMATEC Engineering	Jun – Aug 2024
- Designed a Python-based data quality monitoring system using Pandas and SQL, comparing SQL Server data with multiple external sources (GitHub, Jenkins, Jira) via SQLAlchemy and APIs. - Implemented SMTP-based email alerting for real-time data quality notifications and developed a modular, extensible architecture enabling easy integration of new data sources.	

ACADEMIC PROJECTS

- DeepTest Agent — GraphRAG Code Debugging System** — Enterprise-scale AI testing agent for automated bug detection and test-case generation. Combined FAISS vector search with Neo4j knowledge graphs (GraphRAG) to model code structure, driven by a LangGraph LLM agent with secure Docker-sandboxed code execution.
- AI-Powered Market Skill Insights & Roadmap** — Automated LinkedIn job-description scraping with LLM-based skill extraction and HDBSCAN clustering to surface the most in-demand skills, then generated personalized upskilling roadmaps tailored to candidate resumes.
- AI-Generated Face Detection** — Fine-tuned a Vision Transformer with LoRA, Optuna hyperparameter optimization, regularization, and LR scheduling; added explainability (XAI) for interpretable predictions — **95% test accuracy**.
- ATS Resume Filtering System** — Resume evaluation system using Gemini LLM and SBERT embeddings to extract skills and experience, ranking candidates via similarity measures (cosine, Jaccard) with personalized feedback.

TECHNICAL SKILLS

Programming	Python, Scala, Java, R, C, PHP
AI / ML	ML, DL, NLP, RL, CNNs, LSTM, Transformers, LLMs, Fine-tuning, RAG, AutoEncoders, Knowledge Graphs
Math & Stats	Probability, Calculus, Linear Algebra, Time Series, Hypothesis Testing, Regression, Estimation
Cloud / DevOps	Azure, Docker, Git, MLOps
Databases	MySQL, MongoDB, Neo4j, ChromaDB, Hive, HDFS
Tools	NumPy, Pandas, Matplotlib, Seaborn, Power BI, Flask, Streamlit, Beautiful Soup, Selenium, n8n, LangChain, LangGraph
Soft Skills	Fast learner, team collaboration, problem solving, curiosity

CERTIFICATIONS & LANGUAGES

Certifications	NLP Specialization (DeepLearning.AI) • Machine Learning (Stanford) • Azure AI Fundamentals 📄 • Azure Data Fundamentals 📄
Languages	Arabic (native) • English (proficient) • French (proficient) • Spanish (basic)